

Unlocking Efficiency in Vendor Payments: A Schuster Retail E-commerce Case Study



Findings from Exploratory Data Analysis for Payment Behavior using Python, k-means clustering, and Random forest classifier using GridSearchCV.
81% Precision on Late Payment Detection.

Presented by -
Smriti Pradhan

Business Problem Statement



Vendor payment delays
impacting efficiency and
financial stability



Need to categorize
customers and predict late
payments



Enable prioritization for
collection teams

Goals

1

Understand
Customer
Payment Behavior

2

Segment
Customers by
Payment Plan

3

Predict Delays

4

Prioritize
Collector Work

Answers Needed

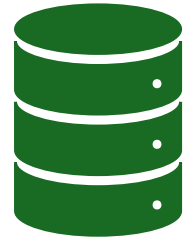
Customer Transaction Analysis

Customer Segmentation Method

Prediction of Delayed Payments

Business Insights from the Model

Model Performance



Precision: Class 0: 48%, Class 1: 81%



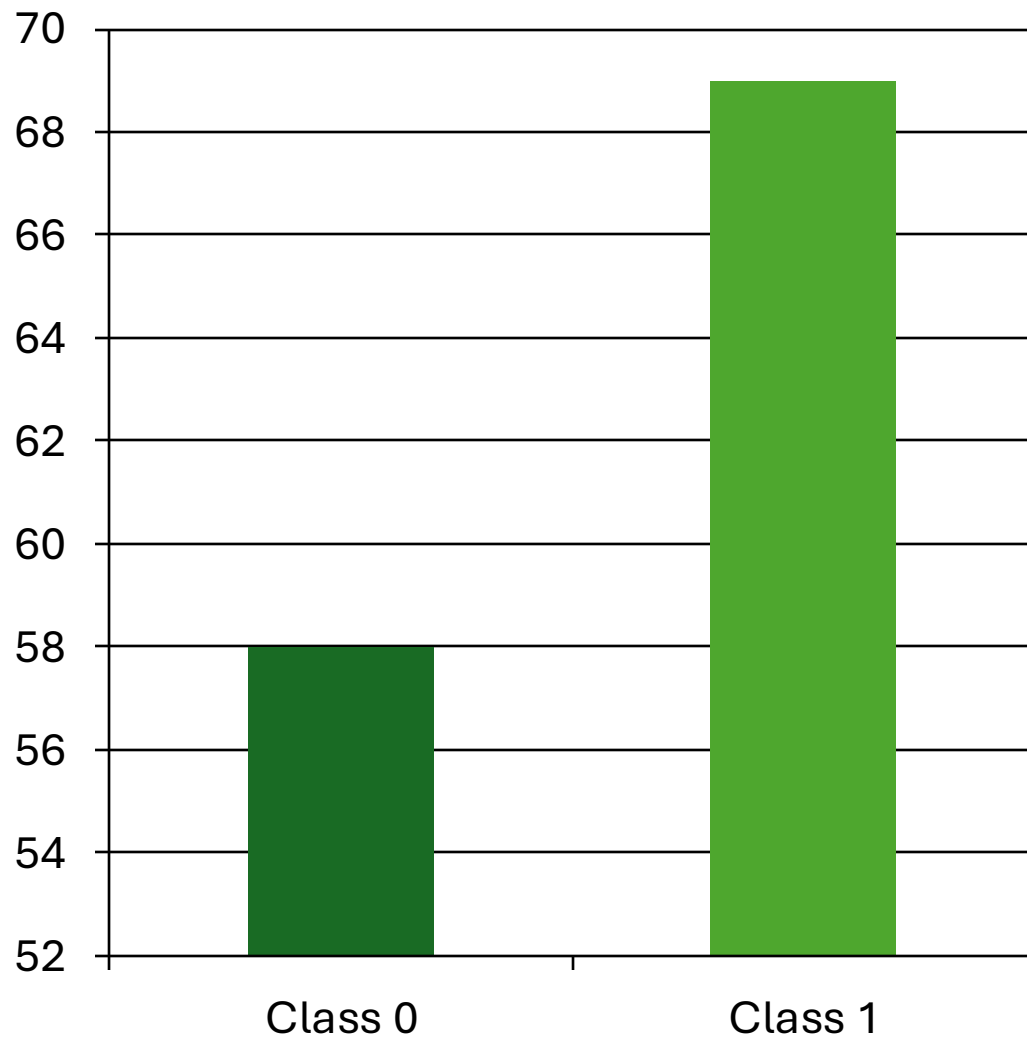
Recall: Class 0: 73%, Class 1: 60%



F1-score: Class 0: 58%, Class 1: 69%

Model
Performance
(F1 Scores)

Series 1



Model Insight #1



Model performs better in predicting late payments than on-time payments.

Hence, we have prioritized analyzing, segmenting, and predicting delayed payments to find key insights.



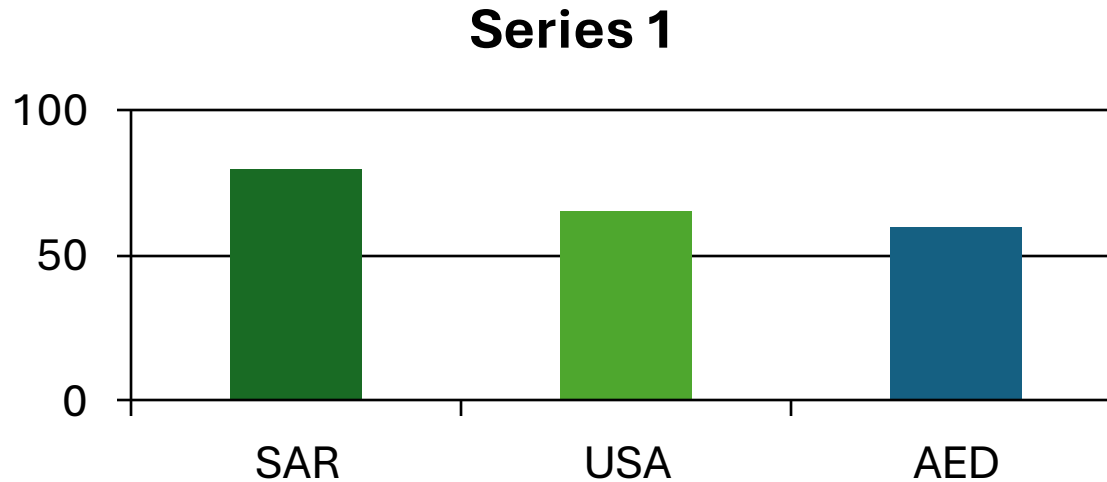
Model performs better
for late payments



Needs improvement
for on-time
predictions

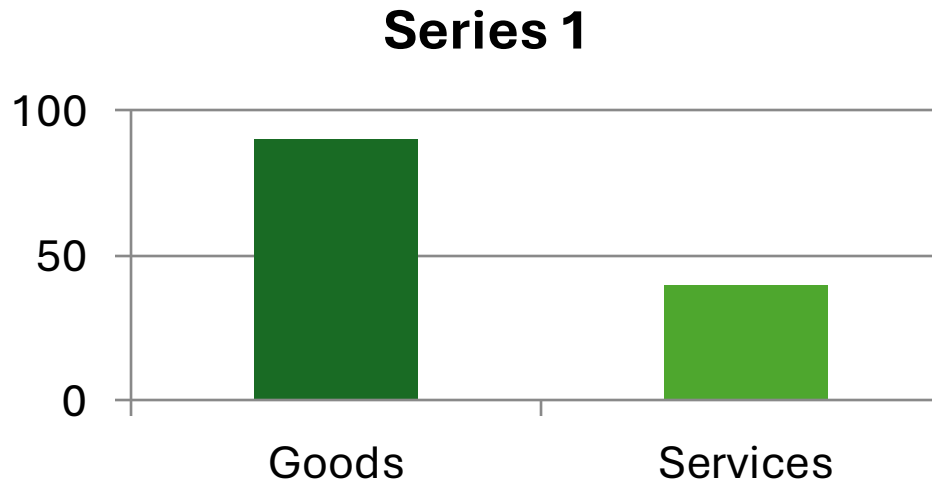
Understanding Customer Payment Behavior

Late Payments by Currency



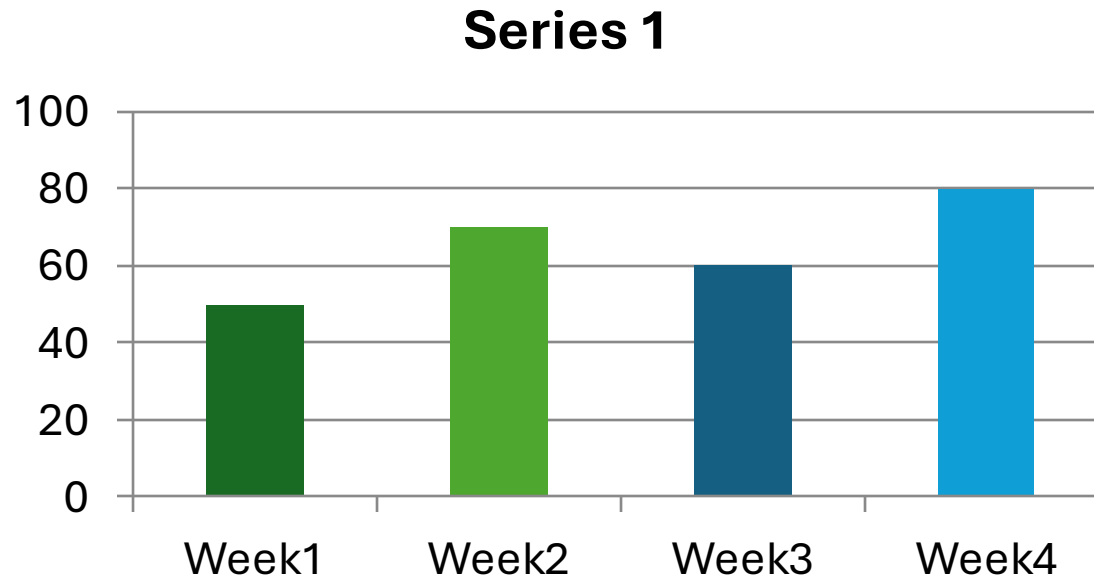
- SAR, USA, and AED currencies contribute most to late payments.
- Recommendation: Prioritize attention and follow-up strategies.

Late Payments by Category



- Goods are the Primary Contributors to Late Payments.
- Concentrate collection strategies specifically on goods invoices.

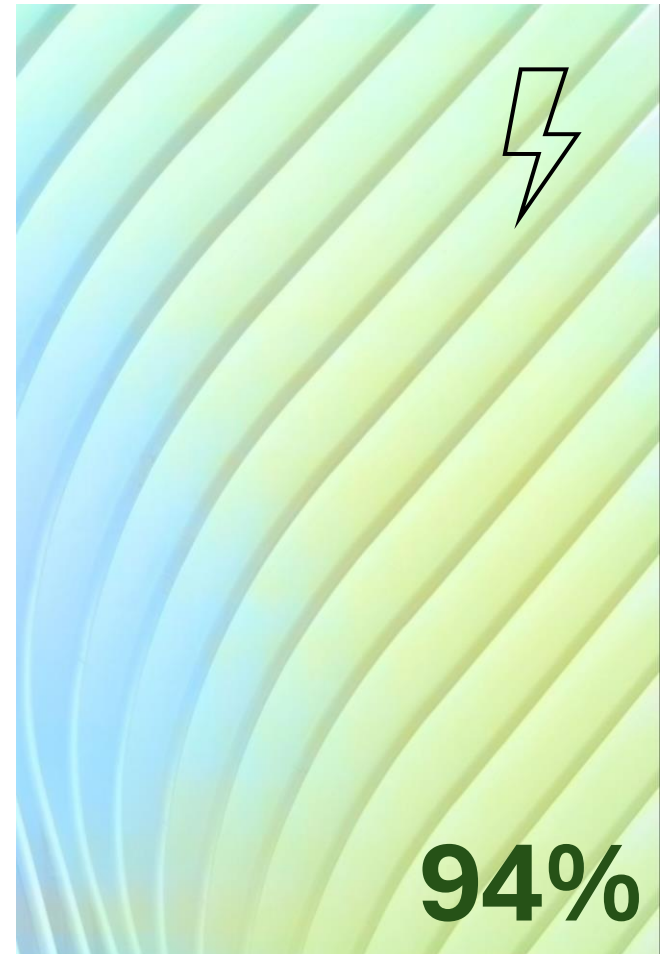
Weekly Seasonality of Late Payments



- Late payments exhibit weekly seasonality
- Teams can proactively prepare and align strategies

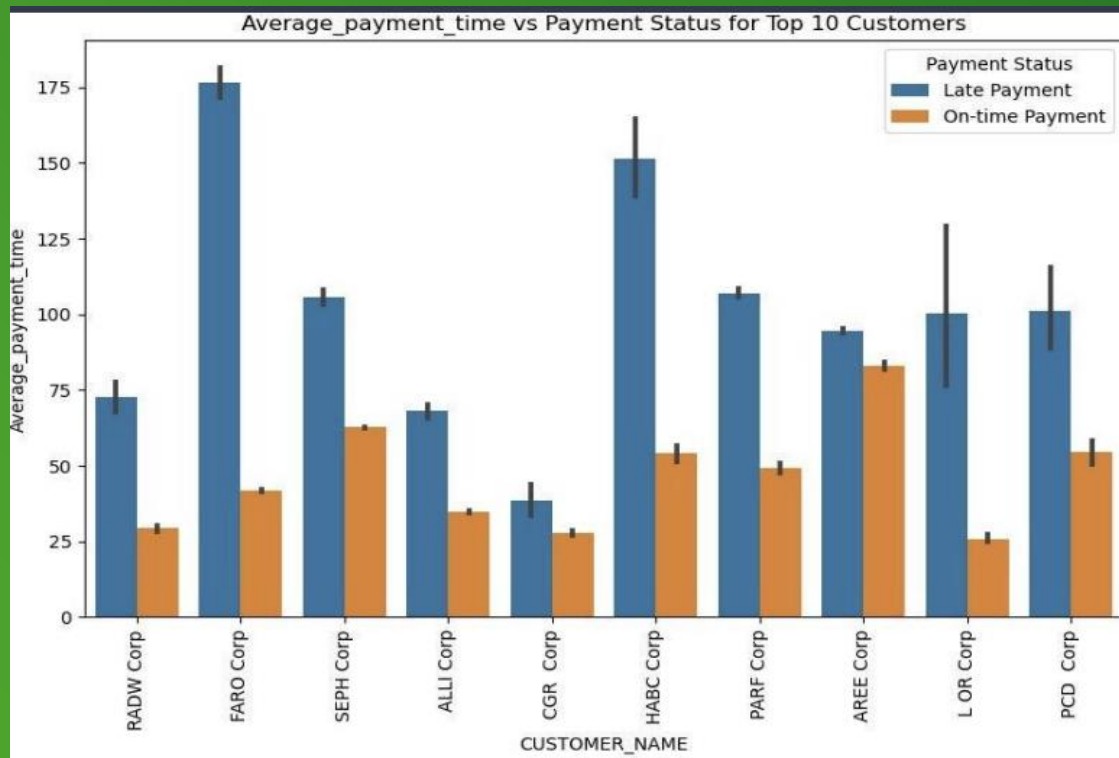
Key Business Insight

Top 10 customers hold 94% of total payment receivables signifying high concentration risk.



Top 10 Customers Compliance

How compliant are they with respect to agreed payment terms?



What can help?

Focused Action

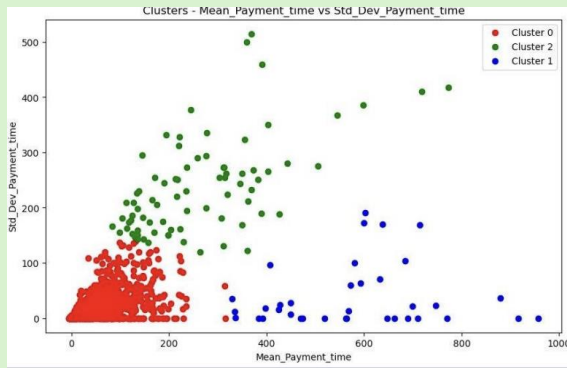
Personalized Approach

Proactive Engagement

Consistent Follow-ups

Data-Driven Adaptation

Customer Segmentation

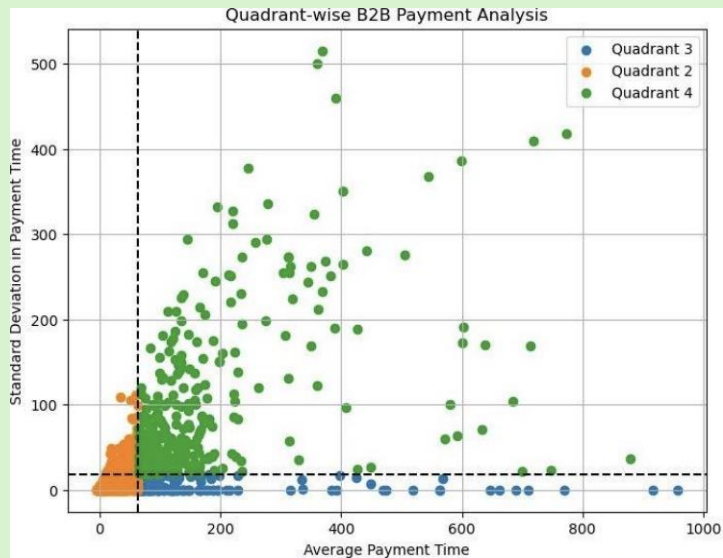


Cluster 0: Punctual Payers

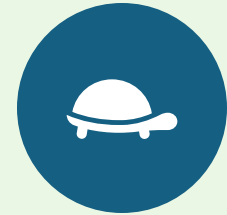
Cluster 1: Erratic Payers

Cluster 2: Variable Payers

Impairment Strategies



PUNCTUAL PAYERS -
RELIABLE CHAMPIONS

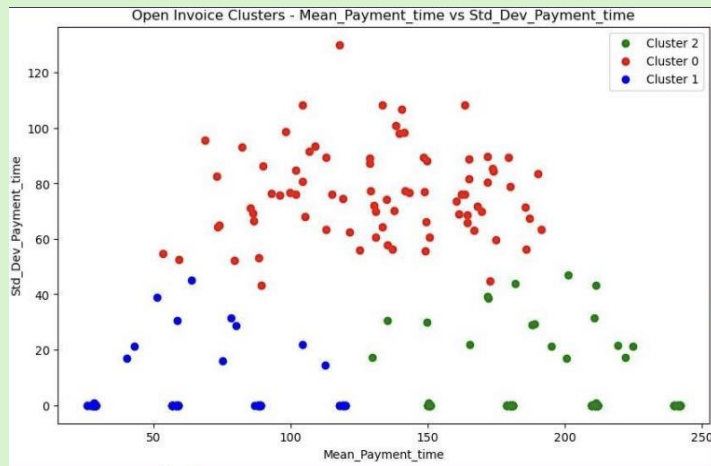


ERRATIC PAYERS -
STEADY IMPROVEMENT
FOCUS



VARIABLE PAYERS -
ADAPTIVE
ENGAGEMENT

Strategies for Open Invoices



PUNCTUAL PAYERS -
RELIABLE CHAMPIONS



ERRATIC PAYERS -
STEADY IMPROVEMENT
FOCUS



VARIABLE PAYERS -
ADAPTIVE
ENGAGEMENT

Recommendations for Getting Customers on the 'Nice List'



Early Payment Discounts



Reward Programs



Referral Incentives

Recommendations for Getting Customers on the 'Nice List'



Exclusive Access



Personalized Gifts/Perks



Recognition



Thank You